

Supersymmetric quantum field theory of $dim = 0, 1$

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1 Introduction

My report is based on David Skinner's lecture notes, David Tong's lecture notes, as well as the book 'Mirror Symmetry' written by Kentaro Hori, Sheldon Katz, and others.

In section 1, I'll give a quick introduction to the terms occurring in the title. In section 2, the $dim = 0$ case will be treated, and in section 3, the $dim = 1$ case will be treated.

1.1 Quantum field theory

Quantum Field Theory is the basic theoretical framework for describing elementary particles and their interactions (excluding gravity), it also plays a major role in areas of physics and mathematics. Next, I will quickly introduce quantum field theory from two aspects: how to choose a quantum field theory and what we will calculate.

We first pick the space on which a QFT lives. It can be a smooth Riemannian (or pseudo-Riemannian) manifold (M, g) of dimension $dim(M) = d$. Moreover, it also be a Riemann surface, where only the conformal class of the metric needed to be specified. After deciding the universe, our next choice is to pick which objects we wish to study, that is to say, the fields. The simplest choice is real-valued (complex-valued) functions on M , or it can be

maps from M to N , where (N, G) is another Riemannian manifold. There are also many choices, each corresponding to a different theory. The next ingredient we need is to specify the action for our theory, which is a function on the space of fields.

If M is closed and compact, the most important object to compute in any QFT is the partition function

$$Z = \int_{\mathcal{C}} [\mathcal{D}\phi] \exp\left(-\frac{S[\phi]}{\hbar}\right) \quad (1)$$

where $\mathcal{C}$ is the infinite dimensional space of fields, $[\mathcal{D}\phi] \exp\left(-\frac{S[\phi]}{\hbar}\right)$ is some sort of measure. Besides partition function, correlation functions in a QFT are also important quantities we wish to compute. These are path integrals with further insertions, of the general form

$$\int_{\mathcal{C}} [\mathcal{D}\phi] \exp\left(-\frac{S[\phi]}{\hbar}\right) \prod_{i=1}^n \mathcal{O}_i[\phi] \quad (2)$$

which will be detailed in the following chapters.

1.2 Supersymmetry

Formally, in any quantum theory containing fermions, the Hilbert space $\mathcal{H}$ splits as $\mathcal{H} = \mathcal{H}_B \oplus \mathcal{H}_F$, where $\mathcal{H}_B$ is the bosonic and $\mathcal{H}_F$ the fermionic Hilbert spaces. The theory is supersymmetric if it contains an operator Q , acting as $Q : \mathcal{H}_B \rightarrow \mathcal{H}_F$ and $Q : \mathcal{H}_F \rightarrow \mathcal{H}_B$, and such that $Q^2 = 0, \{Q, Q^\dagger\} = 2H$.

For much of its development, the main hope for supersymmetry was that it would provide a phenomenologically relevant extension to physics beyond the Standard Model. However, current experimental equipment cannot reach the energy scale where supersymmetric phenomena may occur. Instead, supersymmetry is interesting because it allows us to better understand quantum field theory. For example, the calculation of partition function $\int_{\mathcal{C}} [\mathcal{D}\phi] \exp\left(-\frac{S[\phi]}{\hbar}\right)$ is impossibly difficult, since it's an integral over

an infinite space, and even the definition of it is a difficult problem. Typically, the best we can do is perturbation theory. That is, we pick a particular field configuraion ϕ_0 that solves the classical field equations, and then write down Feynman diagrams describing particles traveling on this background. However, in supersymmetric theories we can often obtain exact results, at least for some observables. By studying them, we understand QFT itself better, even if our final goal is to do away with supersymmetry.

2 Zero dimension

2.1 Fermions and super vector spaces

To consider a supersymmetric theory, we need the notion of Grassmann variables. The definitions and properties are listed below.

1. A $\mathbb{Z}_2$ - *graded* vector space is a vector space $V = V_0 \oplus V_1$ over $\mathbb{R}$ or $\mathbb{C}$ endowed with a parity operation. For a homogeneous vector v , the parity is defined by $|v| = i$, if $v \in V_i$. A basic example is $V = \mathbb{R}^{p|q}$, where V_0 is of dimension p , V_1 of dimension q .
2. Usual operations on graded vector spaces can be introduced, like dual, direct sum, tensor product and so on.
3. What makes V a super vector space is an unusual choice of symmetry operator, defined by

$$s : V \otimes W \rightarrow W \otimes V, s : v \otimes w \mapsto (-1)^{|v||w|} w \otimes v \quad (3)$$

for any homogeneous elements v, w , and extended by linearity to all of V and W . In other words, interchanging the order of any two fermions gives a minus sign. Thus, $Sym^* V_0$ is ordinary, while $Sym^* V_1 \cong \wedge^* V_1$.

4. Another related notion is superalgebra, which is a super vector space $\mathcal{A}$ with a bilinear multiplication map

$$\cdot : \mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A} \quad (4)$$

such that, for any $a, b \in \mathcal{A}$, $|a \cdot b| = |a| + |b|$. It's called commutative if $ab = (-1)^{|a||b|}ba$. For example, to make $\mathbb{R}^{p|q}$ a superalgebra, we have the relations

$$x^i x^j = x^j x^i, \quad x^i \psi^a = \psi^a x^i, \quad \psi^a \psi^b = -\psi^b \psi^a \quad (5)$$

where $x^i \in \mathbb{R}^{p|0}$ are bosonic, $\psi^a \in \mathbb{R}^{0|q}$ are fermionic.

5. The ring of polynomials on V is $\mathcal{O}(V) = \text{Sym}^* V^* \cong \text{Sym}^* V_0^* \otimes \wedge^* V_1^*$, where in the last term, V_0, V_1 is treated as ordinary vector spaces.

Since we want to do path integral, we also need to define differentiation and integration for fermions.

1. A derivation of a commutative superalgebra $\mathcal{A}$ is a linear map $D : \mathcal{A} \rightarrow \mathcal{A}$, such that $D(ab) = (Da)b + (-1)^{|a||D|}a(Db)$. In our example $\mathbb{R}^{p|q}$, we define $\partial/\partial x^i, \partial/\partial \psi^a$, that obey

$$\frac{\partial}{\partial x^i} x^j = \delta_i^j, \quad \frac{\partial}{\partial x^i} \psi^b = 0, \quad \frac{\partial}{\partial \psi^a} \psi^b = \delta_a^b, \quad \frac{\partial}{\partial \psi^a} x^j = 0 \quad (6)$$

and the derivation property $\frac{\partial}{\partial \psi^a}(\psi^b \psi^c) = \delta_a^b \psi^c - \psi^b \delta_a^c$.

2. A vector field on $\mathbb{R}^{p|q}$ is a derivation

$$X(x, \psi) = X^i(x, \psi) \frac{\partial}{\partial x^i} + \chi^a(x, \psi) \frac{\partial}{\partial \psi^a}, \quad (7)$$

it is even if X^i are even and χ^a are odd, odd if X^i are odd and χ^a are even.

3. Since $\psi^2 = 0$, any function of a single fermionic variable ψ is of the form $f + \rho\psi$. We require our integration measure to be translationally invariant, which is to say, if $\psi' = \psi + \eta$, with some fixed $\eta \in \mathbb{R}^{0|1}$, then $\int \psi' d\psi' = \int \psi d\psi$. Thus,

$$\int 1 d\psi = 0, \quad \int \psi d\psi. \quad (8)$$

Generally, if we have n fermionic variables ψ^a , we have

$$\int \psi^{a_1} \psi^{a_2} \dots \psi^{a_n} d^n \psi = \epsilon^{a_1 a_2 \dots a_n}. \quad (9)$$

If we take a linear transformation $\chi^a = N_b^a \psi^b$, after simple calculation, we get $d^n \chi = \frac{1}{\det(N)} d^n \psi$.

2.2 A purely bosonic theory

In zero-dimensional toy model, the universe M is just a point: $M = \{pt\}$. Then, in the simplest case, a field on M is just a map $X : \{pt\} \rightarrow \mathbb{R}^n$. Moreover, the configuraion space of fields is just $\mathbb{R}^n$, the path integral measure $\mathcal{D}X$ is just standard Lebesgue measure. Thus,

$$Z = \int_{\mathbb{R}^n} \exp(-S(X)/\hbar) d^n X \quad (10)$$

where $S(X)$ is an action, which is just a function on $\mathbb{R}^n$ so that the integral converges. The simplest choice is

$$S(X) = \frac{1}{2} M(X, X) = \frac{1}{2} M_{ab} X^i X^j, \quad (11)$$

which makes the partition function Gaussian integral, and can be calculate exactly. We can also plus a linear term on $S(X)$, which won't increase the difficulty. For a general $S(X)$, we may hope to approximate the partition function by expanding around the classical limit $\hbar \rightarrow 0$. In particular, if $S(X)$ has a unique, isolated minimum at $X_0 \in \mathbb{R}^n$, as $\hbar \rightarrow 0^+$, using steepest descent, we have asymptotically

$$\int_{\mathbb{R}^n} e^{-S(X)/\hbar} d^n X \sim (2\pi\hbar)^{n/2} \frac{\exp(-S(X_0)/\hbar)}{\det(\partial_i \partial_j S|_{X_0})} (1 + A_1 \hbar + A_2 \hbar^2 + \dots), \quad (12)$$

and the different terms in this expansion correspond to different types of Feynman diagrams.

2.3 A purely fermionic theory

Since the action must be bosonic, the smallest number of fermions is two, so it is then

$$S(\psi^1, \psi^2) = A\psi^1\psi^2 \quad (13)$$

for some constant A . By the properties of Grassmann variables, $S^2 = 0$, so the partition function

$$\int e^{-A\psi^1\psi^2/\hbar} d\psi^1 d\psi^2 = \int (1 - \frac{A}{\hbar}\psi^1\psi^2) d^2\psi = A/\hbar. \quad (14)$$

More generally, for $2m$ fermions, and an antisymmetric matrix A_{ab} , the partition function is

$$\int e^{-A_{ab}\psi^a\psi^b/2\hbar} d^{2m}\psi = (-\frac{1}{\hbar})^m P(A), \quad (15)$$

where $P(A) = \frac{1}{2^m m!} \epsilon^{a_1 a_2 \dots a_{2m-1} a_{2m}} A_{a_1 a_2} \dots A_{a_{2m-1} a_{2m}}$.

From above, we note that, if our action only contains fermions, then the expansion of e^{-S} must truncate eventually, which is to say, in $\dim = 0$, purely fermionic theories are simpler than purely bosonic ones, which doesn't carry over to higher dimensions.

2.4 A supersymmetric theory

Now we consider a $\dim = 0$ theory containing a bosonic variable $x \in \mathbb{R}$ and two fermionic variables ψ^1, ψ^2 . A general action takes the form

$$S(x, \psi^1, \psi^2) = V(x) + U(x)\psi^1\psi^2, \quad (16)$$

for some smooth functions U, V of the bosonic fields. For general polynomials U, V , the calculation of partition functions is as intractable as a purely bosonic one, since after figure out the easy terms, it will leave with $\int_{\mathbb{R}} U(x) e^{-V(x)/\hbar}$, and we can only evaluate it perturbatively.

Instead, we consider a special form of action, which is related to supersymmetry. Write $\psi = \psi^1 + i\psi^2$, $\bar{\psi} = \psi^1 - i\psi^2$, where i is the imaginary unit.

Let the action be

$$S(x, \psi, \bar{\psi}) = \frac{1}{2}(\partial W)^2 - \bar{\psi}\psi\partial^2 W. \quad (17)$$

It's easy to show that $S(x, \psi, \bar{\psi})$ is invariant under the flow generated by the fermionic vector fields

$$\mathcal{Q} = \psi \frac{\partial}{\partial x} + \partial W(x) \frac{\partial}{\partial \bar{\psi}} \quad (18)$$

$$\mathcal{Q}^\dagger = \bar{\psi} \frac{\partial}{\partial x} - \partial W(x) \frac{\partial}{\partial \psi}, \quad (19)$$

which means $\mathcal{Q}(S) = \mathcal{Q}^\dagger(S) = 0$.

In this case, the QFT is drastically simpler than generic ones. In fact, the method of steepest descent we introduced above is exact. Next, I'll show two ways to understand this.

Firstly, rescale W to λW for some $\lambda \in \mathbb{R}_+$, and let $S_\lambda(x, \psi, \bar{\psi}) = \frac{\lambda^2}{2}(\partial W)^2 - \lambda\bar{\psi}\psi\partial^2 W$ be the action after rescaling. S_λ is invariant under similarly rescaled supersymmetry transformations generated by $\mathcal{Q}_\lambda = \psi \frac{\partial}{\partial x} + \lambda\partial W \frac{\partial}{\partial \bar{\psi}}$ and $\mathcal{Q}_\lambda^\dagger = \bar{\psi} \frac{\partial}{\partial x} - \lambda\partial W \frac{\partial}{\partial \psi}$.

Now, we want to show, the path integral $I(\lambda) = \frac{1}{\sqrt{2\pi}} \int e^{-S_\lambda} dx d\psi d\bar{\psi}$ is independent of λ :

$$\begin{aligned} \frac{d}{d\lambda} I(\lambda) &= -\frac{1}{\sqrt{2\pi}} \int \frac{\partial S_\lambda}{\partial \lambda} e^{-S_\lambda} dx d\psi d\bar{\psi} \\ &= -\frac{1}{\sqrt{2\pi}} \int (\lambda(\partial W)^2 - \bar{\psi}\psi\partial^2 W) e^{-S_\lambda} dx d\psi d\bar{\psi} \\ &= \frac{1}{\sqrt{2\pi}} \int \mathcal{Q}_\lambda^\dagger(\psi\partial W) e^{-S_\lambda} dx d\psi d\bar{\psi} \\ &= \frac{1}{\sqrt{2\pi}} \int \mathcal{Q}_\lambda^\dagger(\psi\partial W e^{-S_\lambda}) dx d\psi d\bar{\psi} \end{aligned} \quad (20)$$

Acting inside [20](#), the derivative $\partial/\partial\psi$ in $\mathcal{Q}_\lambda^\dagger$ strips off the power of ψ , so this term doesn't survive the integration of ψ . For the term generated by acting $\bar{\psi}\partial/\partial x$, it's a total derivative with respect to x , and since $\partial W e^{-\lambda^2(\partial W)^2/2} \rightarrow 0$ as $|x| \rightarrow \infty$, this term also vanishes. Thus we have

$$\frac{d}{d\lambda} I(\lambda) = 0 \quad (21)$$

Note that, the quantity we really care about is $I(1)$, and after the above argument, we can calculate it by

$$I(1) = \lim_{\lambda \rightarrow \infty} I(\lambda). \quad (22)$$

Note that, as $\lambda \rightarrow \infty$, the bosonic factor $e^{\lambda^2(\partial W)^2/2}$ suppresses all contributions to the integral everywhere except where $\partial W = 0$. Thus, the integral receives contributions only from neighbourhood of the critical points of $W(x)$. This is the key property called localization.

The second way to understand the localization is as follows. Let $\mathcal{U}$ be a neighbourhood of $\partial W = 0$. On $\mathcal{U}^c$, we change variables from $(x, \psi, \bar{\psi})$ to $(y, \chi, \bar{\chi})$, where

$$y = x - \frac{\bar{\psi}\psi}{\partial W} \quad \chi = \psi\sqrt{\partial W} \quad \bar{\chi} = \bar{\psi}. \quad (23)$$

Then we have $dx d^2\psi = \sqrt{\partial W(y)} dy d^2\chi$ and $\mathcal{Q}(y) = \mathcal{Q}^\dagger(y) = 0$. Furthermore,

$$S(y, 0, 0) = \frac{1}{2}(\partial W(y))^2 = \frac{1}{2}(\partial W(x))^2 - \partial^2 W(x)\psi\bar{\psi} = S(x, \psi, \bar{\psi}). \quad (24)$$

Thus,

$$\int_{\mathcal{U}^c} e^{-S(x, \psi, \bar{\psi})} dx d^2\psi = \int_{\mathcal{U}^c} e^{-S(y, 0, 0)} \sqrt{\partial W(y)} dy d^2\chi = 0, \quad (25)$$

where the last equality is from the independence of the integrand of the fermionic variables. And we got the conclusion same as the first viewpoint.

Next, we use the localization property to calculate the integral. Suppose $W(x)$ is some generic polynomial of degree D with $D - 1$ isolated non-degenerate critical points. Near any such critical point x_* , we have $W(x) = W(x_*) + \frac{c_*}{2}(x - x_*)^2 + \dots$, so the action becomes $S(x, \psi, \bar{\psi}) = \frac{c_*^2}{2}(x - x_*)^2 - c_*\bar{\psi}\psi + \dots$. Thus the contribution of this critical point to the integral is

$$\begin{aligned} \frac{1}{\sqrt{2\pi}} \int e^{-c_*^2(x-x_*)^2/2} (-1 + c_*\bar{\psi}\psi) dx d^2\psi &= \frac{c_*}{\sqrt{2\pi}} \int e^{-c_*^2(x-x_*)^2/2} dx \\ &= \frac{c_*}{\sqrt{c_*^2}} = \text{sgn}(\partial^2 W|_*). \end{aligned} \quad (26)$$

Summing over all the critical points, we get:

$$I = \sum_{x_*: \partial W|_* = 0} \text{sgn}(\partial^2 W|_*). \quad (27)$$

If W is a polynomial of odd degree, then $\partial W = 0$ must have an even number of roots, with $\partial^2 W$ being alternately positive and negative at each successive root, thus $I_{\text{odd}} = 0$. For even degree polynomial, $I_{\text{even}} = \pm 1$, where the sign depending on the behavior of W as $|x| \rightarrow \infty$.

2.4.1 Landau-Ginzburg theories

Consider a theory of a complex boson z and two complex fermions ψ_1, ψ_2 . We pick a holomorphic function $W(z)$, and take the action to be

$$S[z, \psi_i] = |W'(z)|^2 + W''(z)\psi_1\psi_2 - \bar{W}''(\bar{z})\bar{\psi}_1\bar{\psi}_2. \quad (28)$$

This action is invariant under the two sets of supersymmetry transformations generated by

$$\mathcal{Q}_1 = \psi_1 \frac{\partial}{\partial z} + \partial \bar{W} \frac{\partial}{\partial \psi_2} \quad \mathcal{Q}_1^\dagger = \bar{\psi}_1 \frac{\partial}{\partial \bar{z}} + \partial W \frac{\partial}{\partial \bar{\psi}_2} \quad (29)$$

$$\mathcal{Q}_2 = \psi_2 \frac{\partial}{\partial z} - \partial \bar{W} \frac{\partial}{\partial \psi_1} \quad \mathcal{Q}_2^\dagger = \bar{\psi}_2 \frac{\partial}{\partial \bar{z}} - \partial W \frac{\partial}{\partial \bar{\psi}_1}. \quad (30)$$

By the holomorphism of W , they obey $\{\mathcal{Q}_i, \mathcal{Q}_j\} = 0, \{\mathcal{Q}_i^\dagger, \mathcal{Q}_j^\dagger\} = 0$.

Away from the neighbourhood of critical points of W , we can change variables $(z, \psi_i) \rightarrow (y, \chi_i)$ as $y = z - \frac{\psi_1}{\psi_2}$, $\chi_i = \sqrt{\partial W} \psi_i$, and write the action as $S[y, 0]$. Then the integrand is independent of the fermions, so vanishes under the Berezin integration.

Let z_* be an isolated non-degenerated critical point of W , then $W(z) = W(z_*) + \frac{\alpha}{2}(z - z_*^2 + \dots$, and the quadratic pieces of the action becomes $S^{(2)}[z, \psi_i] = |\alpha(z - z_*)|^2 + \alpha\psi_1\psi_2 - \bar{\alpha}\bar{\psi}_1\bar{\psi}_2$. Thus we can calculate the

partition function around a critical point as follows:

$$\begin{aligned}
I_{z_*} &= \frac{1}{2\pi} \int e^{-|\alpha(z-z_*)|^2} |\alpha|^2 \psi_1 \psi_2 \bar{\psi}_1 \bar{\psi}_2 d^2 z d^4 \psi \\
&= \frac{1}{2\pi} |\alpha|^2 \int e^{-|\alpha(z-z_*)|^2} dz d\bar{z} \\
&= 1
\end{aligned} \tag{31}$$

From this, we know, the partition function is

$$I = \#(\text{critical points of } W) \tag{32}$$

3 One dimension

3.1 Path integral in QM

In one dimension, there are two connected compact manifolds: the circle S^1 and the interval I . The interval can be parametrized by $t \in [0, T]$, the circle by $t \in [0, T)$, with the identification $t \cong t + T$. The most important example of a field on M is a map $x : M \rightarrow N$ to a Riemannian manifold (N, G) , and the standard choice of action is $S[x] = \int_M [\frac{1}{2} G_{ab} \dot{x}^a \dot{x}^b + V(x)] dt$, where $G_{ab}(x)$ is the pullback to M of the Riemannian metric on N .

The usual way to do QM is to pick a Hilbert space $\mathcal{H} = L^2(N)$ and a Hamiltonian $H = -\frac{\hbar^2}{2} \Delta + V$, where $\Delta := \frac{1}{\sqrt{G}} \frac{\partial}{\partial x^a} (\sqrt{G} G^{ab} \frac{\partial}{\partial x^b})$ is the Laplacian acting on $L^2(N)$. The amplitude for the particle to travel from an initial point $y_0 \in N$ to a final point $y_1 \in N$ in Euclidean time T is given by

$$K_T(y_0, y_1) = \langle y_1 | e^{-HT/\hbar} | y_0 \rangle, \tag{33}$$

which is the heat kernel. In fact, it's a function on $I \times N \times N$ that is the solution of the partial differential equation $\hbar \frac{\partial}{\partial t} K_t(x, y) + H K_t(x, y) = 0$ subject to the initial condition $K_0(x, y) = \delta(x - y)$. (This is actually the Schrödinger equation before Wick rotation.)

Assume $V(x) = 0$. Generally, the heat kernel on a Riemannian manifold (N, G) is complicated, but it always has the following asymptotic form:

$$\lim_{\Delta t \rightarrow 0} K_{\Delta t}(x, y) \sim \frac{1}{(2\pi\hbar\Delta t)^{n/2}} a(x) \exp\left(-\frac{d(x, y)^2}{2\hbar\Delta t}\right), \quad (34)$$

where $d(x, y)$ is the distance between x and y measured along a geodesic of the metric, and $a(x)$ is some function constructed from the Riemannian curvature of N .

Following Feynman's intuition, we break the time interval $[0, T]$ into N chunks, each of duration $\Delta t = T/N$, then

$$\langle y_1 | e^{-HT/\hbar} | y_0 \rangle = \int \prod_{i=1}^{N-1} d^n x_i K_{\Delta t}(x_{N-1}, y_1) \cdots K_{\Delta t}(x_1, x_2) K_{\Delta t}(y_0, x_1). \quad (35)$$

Thus, using the asymptotic form 34, we get:

$$\langle y_1 | e^{-HT/\hbar} | y_0 \rangle \sim \frac{1}{(2\pi\hbar\Delta t)^{n/2}} \int \prod_{i=1}^{N-1} \frac{d^n x_i}{(2\pi\hbar\Delta t)^{n/2}} a(x_i) \exp\left[-\frac{1}{\hbar} \sum_{i=0}^N \frac{\Delta t}{2} \left(\frac{d(x_i, x_{i+1})}{\Delta t}\right)^2\right]. \quad (36)$$

Observing this expression, we define loosely a measure on the path space to be

$$\mathcal{D}x = \lim_{N \rightarrow \infty} \left(\frac{1}{2\pi\hbar\Delta t}\right)^{\frac{nN}{2}} \prod_{i=1}^{N-1} d^n x_i a(x_i), \quad (37)$$

which means to integral over values of $x^a(t)$ at each time $t \in [0, T]$. Similarly, since $(d(x_i, x_{i+1})/\Delta t)^2 \rightarrow G_{ab}\dot{x}^a\dot{x}^b$, we may have

$$\lim_{N \rightarrow \infty} \left[\sum_{i=0}^N \frac{\Delta t}{2} \left(\frac{d(x_i, x_{i+1})}{\Delta t}\right)^2\right] = \int_0^T \frac{1}{2} G_{ab}(x) \dot{x}^a \dot{x}^b dt. \quad (38)$$

In fact, in one dimension, while neither the above two expressions have any continuum meaning, they together can give a rigorous definition, called Wiener measure.

Returning to our imprecise definition, note that, the *rhs* in 38 is just the action $S[x]$ with $V(x) = 0$, thus, we obtain the expression we want here:

$$\langle y_1 | e^{-HT/\hbar} | y_0 \rangle = K_T(y_0, y_1) = \int_{C_T[y_0, y_1]} \mathcal{D}x e^{-S[x]/\hbar}, \quad (39)$$

where the integral is taken over the space of paths from y_0 to y_1 in time T .

Next, we'll show how to calculate partition function with path integral:

$$\begin{aligned}\mathcal{Z}(T) &= \text{Tr}_{\mathcal{H}}(e^{-TH}) \\ &= \int d^n y \langle y | e^{-TH} | y \rangle = \int_N d^n y \int_{C_T[y,y]} \mathcal{D}x e^{-S} \\ &= \int_{C_{S^1}} \mathcal{D}x e^{-S/\hbar}\end{aligned}\tag{40}$$

where the last integral is over maps $x : S^1 \rightarrow N$.

As an example, let's consider a free particle with action $S[x] = \int \frac{1}{2} \dot{x}^2 d\tau$ with $x = x + 2\pi R$, which is a bosonic particle living on a circle of radius R . In the canonical picture, the Hamiltonian is $H = -\frac{1}{2} \frac{d^2}{dx^2}$, and the eigenfunctions are $\phi_n(x) = e^{inx/R}$ for $n \in \mathbb{Z}$, so the partition function is

$$\mathcal{Z}(\beta) = \text{Tr}_H(e^{-\beta H}) = \sum_{n \in \mathbb{Z}} e^{\beta n^2 / 2R^2}.\tag{41}$$

Now we calculate it as a path integral over all maps $x : S^1_\beta \rightarrow S^1_{2\pi R}$, weighted by the action. Such maps are classified by winding number $m \in \mathbb{Z}$. So we let

$$x_m(\tau) = y(\tau) + \frac{2\pi m \tau R}{\beta},\tag{42}$$

where y satisfied $y(\tau + \beta) = y(\tau)$ has winding number 0. Then take path integral, the partition function becomes

$$\mathcal{Z}(\beta) = \sum_{m \in \mathbb{Z}} \int e^{-S[x_m]} Dy.\tag{43}$$

At winding number m ,

$$S[x_m] = \frac{2m^2 \pi^2 R^2}{\beta} - \frac{1}{2} \int_{S^1_\beta} y \frac{d^2 y}{d\tau^2} d\tau,\tag{44}$$

decompose the periodic function y into Fourier components and take the path integral measure Dy formally to be $Dy = \prod_{n=1}^{\infty} \frac{dy_n d\tilde{y}_n}{2\pi}$, we get

$$\mathcal{Z}(\beta) = \sum_{m \in \mathbb{Z}} e^{-2\pi^2 m^2 R^2 / \beta} 2\pi R \sqrt{\frac{\beta}{2\pi}} \prod_{n=1}^{\infty} \left(\frac{\beta}{2\pi n}\right)^2.\tag{45}$$

Then we use analytic continuation of Riemann's zeta function to make this expression converging, and finally, we have

$$\mathcal{Z}(\beta) = R \sqrt{\frac{2\pi}{\beta}} \sum_{m \in \mathbb{Z}} e^{-2\pi^2 m^2 R^2 / \beta}. \quad (46)$$

After Poisson resummation, this is equal to the result we calculated with Hamiltonian.

As mentioned in the introduction, we can also use path integral to calculate correlation functions of operators. Since it's not needed in this report, I will not go into detail about it.

3.2 Supersymmetric QM without path integral

3.2.1 Energy spectrum

Recall our setting, we have a Hamiltonian H , and an operator Q , called supercharge, satisfying

$$H = \frac{1}{2} \{Q, Q^\dagger\}, \quad Q^2 = 0. \quad (47)$$

From this, we take the usual expectation value of a state $|\psi\rangle$,

$$2\langle\psi|H|\psi\rangle = \langle\psi|Q^\dagger Q + Q Q^\dagger|\psi\rangle = |Q|\psi\rangle|^2 + |Q^\dagger|\psi\rangle|^2 \geq 0. \quad (48)$$

Thus the energy of any state is necessarily non-negative, and $E = 0 \iff Q|\psi\rangle = Q^\dagger|\psi\rangle = 0$. Moreover, from the definition 47, it's easy to show that $[H, Q] = [H, Q^\dagger]$ also holds.

Now consider the set of states with some fixed energy E , i.e. $H|\psi\rangle = E|\psi\rangle$. If $E \neq 0$, we have

$$\{Q, Q^\dagger\} = 2E. \quad (49)$$

Then define $c = \frac{Q}{\sqrt{2E}}$, it satisfies $\{c, c^\dagger\} = 1$, $c^2 = (c^\dagger)^2 = 0$. Thus the algebra generated by $c, c^\dagger$ has an two-dimensional irreducible representation spanned by $|0\rangle, |1\rangle$, such that $c|0\rangle = 0$, $|1\rangle = c^\dagger|0\rangle$. This means, all energy eigenstates with $E \neq 0$ must come in pairs.

Inspired by the above argument, we define the fermion number operator

$$F = c^\dagger c. \quad (50)$$

It is well defined only on states with energy $E \neq 0$, and obeys $[F, Q] = -Q$, $[F, Q^\dagger] = Q^\dagger$, $[F, H] = 0$, acts as $F|0\rangle = 0$, $F|1\rangle = |1\rangle$.

Correspondingly, the Hilbert space decomposes into "bosonic states" with $F = 0$ and "Fermionic states" with $F = 1$, $\mathcal{H} = \mathcal{H}_B \oplus \mathcal{H}_F$, which is just a $\mathbb{Z}_2$ -graded vector space introduced above.

3.2.2 An example

Consider the quantum mechanics of a particle moving on a line, with an internal degree of freedom, like spin, that can take two different values. The Hilbert space is

$$\mathcal{H} = L^2(\mathbb{R}) \otimes \mathbb{C}^2 = L^2(\mathbb{R})|0\rangle \oplus L^2(\mathbb{R})|1\rangle = \mathcal{H}_B \oplus \mathcal{H}_F, \quad (51)$$

and we may call elements in $\mathcal{H}_F$ fermion, elements in $\mathcal{H}_B$ boson, which are just names.

For supercharge Q , we take

$$Q = (p - ih'(x)) \otimes \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \quad (52)$$

where $p = -id/dx$ and $h(x)$ a real function. Take conjugate,

$$Q^\dagger = (p + ih'(x)) \otimes \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad (53)$$

and so

$$H = \frac{1}{2}\{Q, Q^\dagger\} = \frac{1}{2}(p^2 + (h')^2)\mathbb{1} - \frac{1}{2}h''\sigma^3. \quad (54)$$

The first factor is the Hamiltonian for a particle with unit mass moving on a line with potential $V(x) = \frac{1}{2}(\frac{dh}{dx})^2$ coming with the unit matrix. The second term comes with the Pauli matrix

$$\sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (55)$$

The fermion number operator in this example is

$$F = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad (56)$$

note that it extends to zero energy states for this case.

Next we consider ground states. A general state takes the form $\Psi(x) = (\psi(x), \phi(x))^T$, to qualify as $E = 0$, this must be annihilated by both the supercharges, i.e. $Q\Psi = Q^\dagger\Psi = 0$, meaning that,

$$-i\left(\frac{d}{dx} + h'\right)\psi = 0, \quad -i\left(\frac{d}{dx} - h'\right)\phi = 0. \quad (57)$$

Thus, with supersymmetry, we have split the Schrödinger equation into a pair of decoupled first order differential equations.

3.2.3 The Witten index

The robustness of supersymmetric ground states can be formulated using the Witten index. It's defined by

$$\mathcal{I} = \text{Tr}(-1)^F e^{-\beta H}. \quad (58)$$

From last subsection, we know $\mathcal{H}_B|_{E>0} \cong \mathcal{H}_F|_{E>0}$. This means, the trace over any state with $E > 0$ simply cancels out in the Witten index, thus, the Witten index only receives contributions from the zero energy states. In other words,

$$\mathcal{I} = \dim\mathcal{H}_{0,B} - \dim\mathcal{H}_{0,F}, \quad (59)$$

where $\mathcal{H}_{0,B}$ is the space of $E = 0$ bosonic ground states, and similar for $\mathcal{H}_{0,F}$.

3.3 Path integral for fermions

Consider the quantum mechanics of a complex fermion. Let $\psi, \psi^\dagger$ be fermionic operators obeying $\{\psi, \psi^\dagger\} = 1$. Let the vacuum $|0\rangle$ be defined by

$\psi|0\rangle = 0$. Define the fermionic coherent state $|\eta\rangle = e^{\psi^\dagger \eta}|0\rangle$ and its adjoint $\langle\eta| = \langle 0|e^{\bar{\eta}\psi}$. We can prove $\langle\eta|\eta\rangle = e^{\bar{\eta}\eta}$, $\psi|\eta\rangle = \eta|\eta\rangle$ and

$$1_{\mathcal{H}} = \int e^{-\bar{\eta}\eta} |\eta\rangle\langle\eta| d^2\eta, \quad Tr_{\mathcal{H}}((-1)^F A) = \int e^{-\bar{\eta}\eta} \langle\eta|A|\eta\rangle d^2\eta. \quad (60)$$

We'll use these to construct the path integral for fermions.

$$\begin{aligned} \langle\chi'|e^{-\beta H}|\chi\rangle &= \langle\chi'|e^{-\Delta\tau H}e^{-\Delta\tau H}\dots e^{-\Delta\tau H}|\chi\rangle \\ &= \int \langle\chi'|e^{-\Delta\tau H}|\eta_{N-1}\rangle \dots \langle\eta_2|e^{-\Delta\tau H}|\eta_1\rangle \langle\eta_1|e^{-\Delta\tau H}|\chi\rangle \prod_{k=1}^{N-1} e^{-\bar{\eta}_k\eta_k} d^2\eta_k \end{aligned} \quad (61)$$

Since $|\eta_k\rangle$, $\langle\eta_k|$ are eigenstates of ψ and $\psi^\dagger$ respectively, for an infinitesimal $\Delta\tau$,

$$\langle\eta_k|e^{\Delta\tau H(\psi^\dagger, \psi)}|\eta_k\rangle = e^{-\Delta\tau H(\bar{\eta}_k, \eta_k)} \langle\eta_{k+1}|\eta_k\rangle = e^{-\Delta\tau H(\bar{\eta}_k, \eta_k)} e^{\bar{\eta}_{k+1}\eta_k}. \quad (62)$$

Then the full heat kernel becomes

$$\begin{aligned} \langle\chi'|e^{-\beta H}|\chi\rangle &= \lim_{N\rightarrow\infty} \int \exp\left(\sum_{k=1}^N \bar{\eta}_k\eta_{k-1} - \Delta\tau H(\bar{\eta}_{k-1}, \eta_{k-1})\right) \prod_{k=1}^{N-1} e^{-\bar{\eta}_k\eta_k} d^2\eta_k \\ &= \lim_{N\rightarrow\infty} \int \exp\left(-\sum_{k=1}^N \left[\eta_k \frac{\bar{\eta}_k - \eta_{k-1}}{\Delta\tau} + H(\bar{\eta}_{k-1}, \eta_{k-1})\right] \Delta\tau\right) e^{\bar{\eta}_N\eta_N} \prod_{k=1}^{N-1} d^2\eta_k, \end{aligned} \quad (63)$$

where we set $\eta_0 = \chi$, and $\eta_N = \chi'$. We recognize the contents of the square brackets as a discretization of the Euclidean action

$$S[\eta, \bar{\eta}] = \int_0^\beta [\bar{\eta}\dot{\eta} + H(\bar{\eta}, \eta)] d\tau, \quad (64)$$

and finally we can write the heat kernel for the fermionic system as a path integral

$$\langle\chi'|e^{-\beta H}|\chi\rangle = \int e^{-S[\psi, \psi^\dagger]e^{\psi^\dagger(\beta)\psi(\beta)}} \mathcal{D}\psi \mathcal{D}\psi^\dagger, \quad (65)$$

where we formally integrate over all $\psi(\tau)$ such that $\psi(0) = \chi$, $\psi(\beta) = \chi'$.

The supertrace, i.e. $Tr((-1)^F e^{-\beta H})$, can also be calculated with path integral:

$$Tr((-1)^F e^{-\beta H}) = \int_P e^{-S[\psi, \psi^\dagger]} \mathcal{D}\psi \mathcal{D}\psi^\dagger, \quad (66)$$

where the integral is computed with fermions that are periodic around the circle.

3.4 Witten index revisited

Recall the example in 3.2.2, where we consider a particle moving on a line with spin. In fact this system can be described by the following action:

$$S = \int dt L = \int dt \left[\frac{1}{2} \dot{x}^2 + i\psi^\dagger \dot{\psi} - \frac{1}{2} h'^2 + h'' \psi^\dagger \psi \right], \quad (67)$$

where ψ and its conjugate $\psi^\dagger$ are Grassmann variables.

In terms of the fermionic degrees of freedom, the supercharges Q and $Q^\dagger$ take the form

$$Q = (p - ih'\psi), \quad Q^\dagger = (p + ih'\psi)\psi^\dagger. \quad (68)$$

Next, we calculate the Witten index of this system. By the formula in the last subsection, we focus on Euclidean time as well as periodic path.

$$S_E[x, \psi, \psi^\dagger] = \oint d\tau \left[\frac{1}{2} \left(\frac{dx}{d\tau} \right)^2 + \psi^\dagger \frac{d\psi}{d\tau} + \frac{1}{2} h' - h'' \psi^\dagger \psi \right] \quad (69)$$

where $\oint$ is to remind us that the path is periodic. The Euclidean action is invariant under the Wick rotated supersymmetry transformations:

$$\delta x = \epsilon^\dagger \psi - \epsilon \psi^\dagger, \quad \delta \psi = \epsilon \left(\frac{dx}{d\tau} + h' \right), \quad \delta \psi' = \epsilon^\dagger \left(-\frac{dx}{d\tau} + h' \right). \quad (70)$$

And the Witten index is

$$\mathcal{I} = Tr((-1)^F e^{\beta H}) = \int \mathcal{D}x \mathcal{D}\psi^\dagger \mathcal{D}\psi e^{S_E[x, \psi, \psi^\dagger]}. \quad (71)$$

Like $dim = 0$ case, we want to show, after rescale $h \rightarrow \lambda h$, where $\lambda > 0$, the Witten index doesn't change. Differentiate with respect to λ :

$$\frac{d\mathcal{I}}{d\lambda} = \int \mathcal{D}x \mathcal{D}\psi^\dagger \mathcal{D}\psi \left(-\oint d\tau (\lambda h'^2 - h'' \psi^\dagger \psi) \right) e^{-S_E}. \quad (72)$$

Note that, after rescaling, one of the supercharges becomes

$$Q_\lambda^\dagger = \int dt [-\psi^\dagger(t) \frac{\delta}{\delta x(t)} + (\frac{dx}{d\tau} + \lambda h') \frac{\delta}{\delta \psi(t)}]. \quad (73)$$

After calculation and some argument like that in the $dim = 0$ case, we can show that

$$\frac{d\mathcal{I}}{d\lambda} = \int \mathcal{D}x \mathcal{D}\psi^\dagger \mathcal{D}\psi Q_\lambda^\dagger (-e^{-S_E} \oint d\tau h' \psi) = 0. \quad (74)$$

Thus we can let $\lambda \rightarrow \infty$ to calculate $\mathcal{I}$. Here the potential term in the action suppresses all contribution except for the finite number of constant maps $x(\tau) = X$, such that $h'(X) = 0$.

All of this means that the path integral expression for the Witten index is

$$\begin{aligned} Tr(-1)^F e^{-\beta H} &= \int \mathcal{D}x \mathcal{D}\psi^\dagger \mathcal{D}\psi e^{-S_E} \\ &= \sum_X \frac{\det(d/d\tau - h''(X))}{\det^{1/2}(-d^2/d\tau^2 + h''(X)^2)} \\ &= \sum_X \frac{-h''(X)}{|h''(X)|} \\ &= \sum_X \text{sign}(-h''(X)). \end{aligned} \quad (75)$$

In order to obtain more interesting results, we have to consider quantum systems with closer connections to geometry. For example, we may consider nonlinear sigma models, and after calculate the Witten index in two different ways, we will prove the Atiyah-Singer index theorem for the Dirac operator on an even-dimensional manifold.